Developer Guide: Nano Banana 2 with the Gemini Interactions API

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Nano Banana 2 (**gemini-3.1-flash-image-preview**) is our latest image generation model. It runs at Gemini Flash speed with the capabilities of Nano Banana Pro's world knowledge, precise text rendering, and subject consistency across up to 14 reference images. It comes with a new **Grounding with Google Image Search** feature that lets the model retrieve real images from the web and use them as visual context during generation.

This Developer Guide uses the Gemini Interactions API, our next-generation unified interface for Gemini, to build a **personalized Japan travel brochure**. In 4 code snippets, you'll go from a simple text prompt to a search-grounded, personalized image, compositing a real person into a scene of Kyoto, informed by Google Search results.

1. **Text → Image:** Pure creative generation
2. **+ Google Search (web):** Real-world factual grounding
3. **+ Google Search (web + images):** Visual accuracy from retrieved images
4. **+ Reference photo:** Subject-consistent personalization

```bash
pip install google-genai
export GEMINI_API_KEY="your-api-key"
```

You can create and manage all your Gemini API Keys from Google AI Studio.

1. Text to Image

Describe what you want, get an image back. Nano Banana 2 handles photorealism, text rendering, and complex compositions without any tools.

```python
```

```
import base64
from google import genai

client = genai.Client()

interaction = client.interactions.create(
    model="gemini-3.1-flash-image-preview",
    input=f"""A cinematic travel poster for Kyoto, Japan in autumn.
Cherry-red maple leaves
frame a traditional wooden torii gate at golden hour. The text 'KYOTO' is
rendered in
elegant serif typography at the bottom. Photorealistic, warm tones, 35mm
film look.""",
)

images = [output for output in interaction.outputs if output.type ==
"image"]
with open("01_kyoto_poster.png", "wb") as f:
    f.write(base64.b64decode(images[0].data))
```

KYOTO

2. Text to Image with Google Search (Web Search)

Adding a **google_search** tool makes the model search the web for factual information before generating.

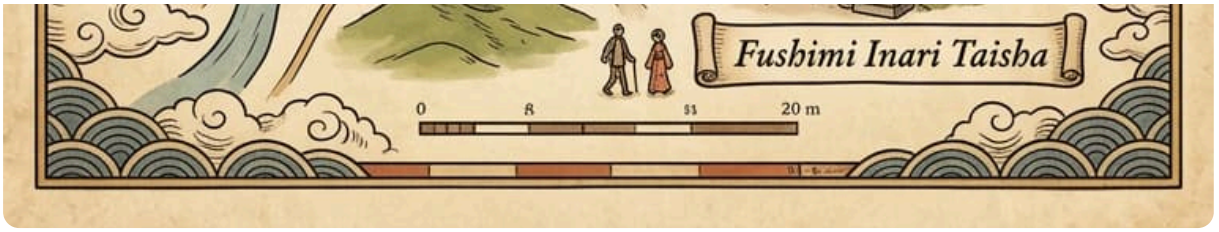
```python
import base64
from google import genai

client = genai.Client()

interaction = client.interactions.create(
    model="gemini-3.1-flash-image-preview",
    input=f"""Search for the top 5 landmarks to visit in Kyoto, Japan.
Then create a beautiful
illustrated travel map poster showing these landmarks with their real
names labeled. Use a
vintage hand-drawn cartography style with warm earth tones. Title it
'KYOTO WALKING GUIDE'.""",
    tools=[
        {
            "type": "google_search",
            "search_types": ["web_search"]
        }
    ]
)

images = [output for output in interaction.outputs if output.type ==
"image"]
with open("02_kyoto_map.png", "wb") as f:
    f.write(base64.b64decode(images[0].data))
```

KYOTO
WALKING GUIDE
N
W
E
S
Katsura
Kinkaku-ji
(Golden Pavilion)
Arashiyama
Bamboo Grove
Kiyomizu-dera
Temple
Gion District
Katsura
Kamo
Katsura



3. Text to Image with Google Search (Web + Image Search)

Nano Banana 2 adds Image Search as a new search type alongside Web Search, controlled via the **search_types** parameter. The model retrieves actual photographs from Google Image Search and uses them as visual references during generation.

With text-only search, the model knows Fushimi Inari has orange torii gates. With Image Search, it sees exactly what they look like.

```python
import base64
from google import genai

client = genai.Client()

interaction = client.interactions.create(
    model="gemini-3.1-flash-image-preview",
    input=f"""Search for what Fushimi Inari Shrine in Kyoto looks like.
Then generate a stunning
photorealistic travel poster of Fushimi Inari's famous orange torii gate
tunnel at sunset with
a light rain falling. Include the text 'FUSHIMI INARI' in a clean, modern
sans-serif font at
the bottom.""",
    tools=[
        {
            "type": "google_search",
            "search_types": ["web_search", "image_search"]
        }
    ]
)

images = [output for output in interaction.outputs if output.type ==
"image"]
with open("03_fushimi_inari.png", "wb") as f:
    f.write(base64.b64decode(images[0].data))
```

FUSHIMI INARI

You can also use **image_search** on its own without **web_search**:

```python
tools=[{"type": "google_search", "search_types": ["image_search"]}]
```

4. Text + Image to Image with Google Search (Web + Image Search)

We can combine a **reference photo** of a person with text instructions and search grounding. Nano Banana 2 preserves the person's appearance from the reference image in the generated output.

We provide a headshot and ask the model to place the person into the Arashiyama bamboo grove, with Image Search ensuring the scene matches the real location.

```python
import base64
from google import genai

client = genai.Client()

image_data = base64.b64encode(open("assets/headshot.png",
"rb").read()).decode()

interaction = client.interactions.create(
    model="gemini-3.1-flash-image-preview",
    input=[
        {
            "type": "text",
            "text": f"""Search for what the Arashiyama bamboo grove in
Kyoto looks like. Then
generate a photo of this person (from the provided image) standing in the
Arashiyama bamboo
```

```
grove during golden hour. They are wearing a casual travel outfit and
looking up at the
towering bamboo stalks with a sense of wonder. Photorealistic, cinematic
lighting, shallow
depth of field."""
        },
        {
            "type": "image",
            "data": image_data,
            "mime_type": "image/png"
        }
    ],
    tools=[
        {
            "type": "google_search",
            "search_types": ["web_search", "image_search"]
        }
    ]
)

images = [output for output in interaction.outputs if output.type ==
"image"]
with open("04_arashiyama_portrait.png", "wb") as f:
    f.write(base64.b64decode(images[0].data))
```

This single API call combines three Nano Banana 2 capabilities:

- **Subject consistency:** Preserves the person's appearance from the reference photo
- **Image Search grounding:** Retrieves real images of the bamboo grove for visual accuracy
- **Web Search grounding:** Pulls factual context about the location

Tips & Best Practices

- The same prompt produces different images on each run.
- Some prompts may be blocked by safety filters. If your prompt is rejected, try adjusting the description.
- The **search_types** parameter with **image_search** requires the **gemini-3.1-flash-image-preview** model specifically.
- Subject consistency works best with clear, well-lit headshots. The model can maintain resemblance for up to 5 characters and 14 reference objects.

Other Resources

- Nano Banana 2 Blog Post
- Image Generation Documentation
- Interactions API Documentation
- Try in AI Studio
- Pricing